

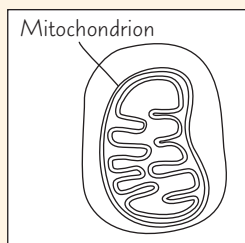
9 Cellular Respiration and Fermentation

KEY CONCEPTS

- 9.1** Catabolic pathways yield energy by oxidizing organic fuels *p. 165*
- 9.2** Glycolysis harvests chemical energy by oxidizing glucose to pyruvate *p. 170*
- 9.3** After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules *p. 171*
- 9.4** During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis *p. 174*
- 9.5** Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen *p. 179*
- 9.6** Glycolysis and the citric acid cycle connect to many other metabolic pathways *p. 182*

Study Tip

Make a visual study guide: Draw a cell with a large mitochondrion, labeling the parts of the mitochondrion. As you go through the chapter, add key reactions for each stage of cellular respiration, linking the stages together. Label the carbon molecule(s) with the most energy and the carbon molecule(s) with the least energy. Your cell can be a simple sketch, as shown here.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 9
- BioFlix® Animation: Cellular Respiration
- Figure 9.12 Walkthrough: Free-Energy Change During Electron Transport

For Instructors to Assign (in Item Library)

- BioFlix® Tutorial: Glycolysis
- BioFlix® Tutorial: Cellular Respiration: Inputs and Outputs

Ready-to-Go Teaching Module

(in Instructor Resources)

- Oxidative Phosphorylation (Concept 9.4)



Figure 9.1 This hoary marmot (*Marmota caligata*) obtains energy for its cells by feeding on plants. In the process of cellular respiration, mitochondria in the cells of animals, plants, and other organisms break down organic molecules, generating ATP and waste products: carbon dioxide, water, and heat. Note that energy flows one way, but chemicals are recycled.

How is the chemical energy stored in food used to generate ATP, the molecule that drives most cellular work?

